

BCSE498J Project-II – Capstone Project
CRACK DETECTION IN INFRASTRUCTURE
USING OPTIMIZED YOLO MODELS

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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DECLARATION

I hereby declare that the project entitled **CRACK DETECTION IN INFRASTRUCTURE USING OPTIMIZED YOLO MODELS** submitted by me, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* to VIT is a record of bonafide work carried out by me under the supervision of **Dr. Anindita Kundu.**

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place : Vellore

Date :

Signature of the Candidate

CERTIFICATE

This is to certify that the project entitled **CRACK DETECTION IN INFRASTRUCTURE USING OPTIMIZED YOLO MODELS** submitted by **Rudraksh Agarwal (21BCT0005)**, School of Computer Science and Engineering, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by him under my supervision during Winter Semester 2024-2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place : Vellore

Date :

Signature of the Guide

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EXECUTIVE SUMMARY

Structural Health Monitoring (SHM) is essential for maintaining the safety, reliability, and longevity of buildings and bridges. Surface crack detection is a critical aspect of preventive maintenance, especially for large-scale infrastructures and historical structures, where undetected cracks can escalate into catastrophic failures. The 2018 collapse of the Morandi Bridge in Genoa, Italy—which tragically claimed 43 lives—highlights the devastating consequences of neglected structural damage. Similarly, the 2021 Champlain Towers South condominium collapse in Surfside, Florida, resulted in 98 fatalities due to unnoticed structural deterioration. In India, the 2016 Kolkata flyover collapse and the 2019 Mumbai footbridge collapse also underscore the severe risks associated with inadequate infrastructure monitoring. Traditional manual inspection methods are time-consuming, labour-intensive, and prone to human error, emphasizing the need for efficient, automated crack detection systems that can provide timely and accurate assessments.

This project aims to develop an optimal automated crack detection system using the advanced You Only Look Once (YOLO) family of models—YOLOv8, YOLOv9, YOLOv10, and the latest YOLOv11. These models are widely recognized for their high-speed, real-time object detection capabilities, making them ideal for rapid damage assessment. The models will be trained on diverse, high-resolution datasets that include various crack patterns under real-world conditions, such as varying lighting, occlusions, complex backgrounds, and different crack sizes and orientations. Transfer learning will further enhance the models to adapt to practical deployment scenarios.

To ensure accurate comparisons, two powerful optimization algorithms—Stochastic Gradient Descent (SGD) and AdamW—will be implemented and evaluated across all YOLO versions. SGD is efficient in handling large-scale data and is widely used in deep learning models for its simplicity and stability. AdamW, on the other hand, improves model convergence and generalization by decoupling weight decay from gradient updates. The models will be assessed based on detection accuracy, inference speed, and mean average precision (mAP) to determine the optimal configuration that balances precision and efficiency for real-time applications.

The optimized YOLO model can also be integrated with Unmanned Aerial Vehicles (UAVs) for rapid and remote crack detection in inaccessible areas. This system enhances structural health monitoring by providing a scalable and effective solution for real-time crack detection in buildings and bridges. By advancing automated inspection technologies, this project contributes to proactive maintenance, minimizing the risk of catastrophic failures, and ensuring the long-term resilience of critical infrastructure.

Keywords - Machine Learning, Neural Networks, Anomaly Detection, YOLO, ADAMW, SGD.

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List of Abbreviations

AdamW	Adaptive Moment Estimation with Weight Decay
AI	Artificial Intelligence
AIoU	Adaptive Intersection over Union
API	Application Programming Interface
ARB	Auxiliary Reversible Branch
AWS	Amazon Web Services
CBLinear	Cross-Branch Linear Fusion
CBS	Convolutional, Batch Normalization, and SiLU activation
CFD	Crack Forest Dataset
CIoU	Complete Intersection over Union
Colab	Google Colaboratory
CPU	Central Processing Unit
C2f	Cross Stage Partial Network with Efficient Layer Aggregation Network
CNN	Convolutional Neural Network
CSP	Cross Stage Partial
CSPDarknet	Cross Stage Partial Darknet (YOLO backbone architecture)
CSPNet	Cross Stage Partial Network
DL	Deep Learning
DJI	Da-Jiang Innovations
ELAN	Efficient Layer Aggregation Networks
F1-Score	Harmonic Mean of Precision and Recall
FLOPs	Floating Point Operations per Second
FPS	Frames Per Second

FPN	Feature Pyramid Network
GELAN	Generalized Efficient Layer Aggregation Network
GitHub	Git Repository Hosting Platform
GPU	Graphics Processing Unit
HD	High Definition
IoT	Internet of Things
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
mAP	Mean Average Precision
ML	Machine Learning
NAS	Neural Architecture Search
NDT	Non-Destructive Testing
ONNX	Open Neural Network Exchange
OpenCV	Open Source Computer Vision Library
PAN	Path Aggregation Network
PANet	Path Aggregation Network
PGI	Programmable Gradient Information
PyTorch	Python-based Machine Learning Library
RAM	Random Access Memory
R-CNN	Region-based Convolutional Neural Network
ReLU	Rectified Linear Unit
RepNcSPELAN4	Reparameterized Non-convolutional Split Path Efficient Layer Aggregation Network v4
Roboflow	Dataset preparation and deployment platform for computer vision
RT-DETR	Real-Time DEtection TRansformer
SDNET2018	Structural Damage NETwork 2018 Dataset

SGD	Stochastic Gradient Descent
SHM	Structural Health Monitoring
SiLU	Sigmoid Linear Unit
SPPF	Spatial Pyramid Pooling – Fast
TensorRT	NVIDIA Tensor Runtime
UAV	Unmanned Aerial Vehicle
UI	User Interface
YOLO	You Only Look Once
YOLOv1–v10	YOLO Version 1 to Version 10

Symbols and Notations

w	Weights of the neural network
η	Learning rate
$\nabla L(w)$	Gradient of the loss function with respect to weights
$\hat{m}$ (m-hat)	Bias-corrected first moment estimate (mean of gradients)
$\hat{v}$ (v-hat)	Bias-corrected second moment estimate (uncentered variance of gradients)
ε	A small constant to prevent division by zero (e.g., $1e-8$)
λ	Weight decay factor (used for regularization in AdamW)